AI Powered Stock Market Insight Generator for the Colombo Stock Exchange

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Retail investors on the Colombo Stock Exchange often receive charts, prices, and technical indicators without a clear explanation of what is driving a movement or whether an unusual observation is reliable. This thesis develops the market modelling and explainable artificial intelligence component of an integrated stock market insight generator. The component collects historical CSE price and volume data, prepares time-series features, learns expected market behaviour, identifies unusual deviations, and explains important feature contributions and interactions. A modular research and development process was used, combining data cleaning, feature engineering, forecasting, anomaly-aware analysis, explainability analysis, integration testing, and evaluation against chronological historical data. The prototype is designed to provide price-path estimates, anomaly evidence, factor contributions, consistency indicators, and plain-language limitations rather than unsupported trading recommendations. The resulting component is integrated with financial-report, news and sentiment, and external-risk modules through the project API. The work demonstrates that a locally focused and explainable workflow can make quantitative market analysis easier to inspect, while also showing that forecast uncertainty, data quality, changing market conditions, and association-versus-causation limitations must remain visible to users.

Keywords: stock market modelling, explainable AI, anomaly detection, Colombo Stock Exchange, investor insight

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - Application Programming Interface

CSE - Colombo Stock Exchange

JSON - JavaScript Object Notation

LSTM - Long Short-Term Memory

SHAP - SHapley Additive exPlanations

Python - Programming language used for the modelling workflow

1 INTRODUCTION

1.1 Overall project context

The wider research project is an AI-powered stock market insight generator for the Colombo Stock Exchange. It combines four complementary evidence sources: quantitative market behaviour, financial-report understanding, news and sentiment analysis, and external-market risk context. The objective is not to replace an investor or financial professional, but to reduce the difficulty of interpreting several evidence streams at the same time.

The integrated product accepts a selected CSE company and a recent company report, processes the available evidence through independent components, and presents a unified investor-oriented explanation. This structure recognises that stock movements are rarely explained by a single variable and that a useful system must show both what the data suggests and where the evidence is weak.

1.2 Individual component context

This individual thesis covers the Market Modeling and Explainable AI Insight Generation component. The component is responsible for acquiring and validating historical stock observations, preparing modelling features, estimating expected price behaviour, detecting unusual deviations, and explaining the contribution of market factors to the generated output.

The component is implemented as a research-supported runtime that is called by the integrated Express application. Its output includes market history summaries, forecast horizons, favourable and adverse ranges, anomaly observations, factor contributions, model-quality evidence, and limitations that are rendered in the dashboard.

1.3 Problem statement

Retail investors commonly encounter raw prices, volumes, and technical indicators without an accessible explanation of how those variables relate to the observed movement. A prediction by itself does not show whether the latest observation is

unusual, which factors contributed to it, how stable the contribution is over time, or how much uncertainty should be attached to the estimate.

The problem addressed by this component is therefore the design of a market-analysis workflow that produces useful quantitative evidence while remaining explainable, anomaly-aware, and honest about uncertainty for an emerging-market setting such as the CSE.

1.4 Research gap

Many stock prediction studies optimise predictive accuracy using historical data but provide limited explanation of the result. Other systems provide dashboards with indicators but do not separate normal movement from unusual movement or test whether an explanation remains consistent across time. A further gap exists in the practical localisation of these ideas for CSE data, where historical coverage, liquidity, missing observations, and market structure differ from large international datasets.

This work addresses the gap by combining behaviour modelling, anomaly detection, feature attribution, interaction analysis, and temporal consistency checking in one component. The design also presents associations as context rather than as proof that a factor caused the share-price movement.

1.5 Aim and objectives

The aim is to develop and evaluate an explainable and anomaly-aware market modelling component for generating investor-friendly insight from CSE market data.

The objectives are to: (1) collect and validate historical price and volume data; (2) engineer features that represent trend, volatility, liquidity, and market behaviour; (3) develop a reproducible forecasting workflow; (4) detect deviations between expected and observed prices; (5) explain predictions and anomalies using feature contributions and interactions; (6) assess explanation stability over time; (7) expose the output through the integrated platform; and (8) evaluate the component using chronological validation, software tests, and integration checks.

1.6 Research questions or hypotheses

RQ1: Can historical CSE price and volume data be transformed into a reproducible market-behaviour model that provides useful short-horizon estimates?

RQ2: Can anomaly-aware comparison distinguish an unusual price observation from ordinary variation while reporting the supporting evidence?

RQ3: Can feature attribution, interaction analysis, and temporal consistency checks produce explanations that are understandable and appropriately qualified for investors?

RQ4: Can the component integrate with report, news, and external-risk modules without hiding uncertainty or presenting associations as causal conclusions?

1.7 Scope limitations and assumptions

The scope is limited to the supported CSE stocks and the historical data available to the project. The component is not a trading bot, does not place orders, and does not guarantee a future price. Forecast intervals are evidence-based estimates rather than guaranteed best or worst outcomes.

The system assumes that historical observations are sufficiently complete for the selected stock and that the data source can be refreshed. Thin trading, corporate actions, missing dates, structural market changes, external shocks, and limited validation windows can reduce reliability. Explanations describe model associations and contribution patterns; they do not prove that a factor caused a movement.

1.8 Individual contribution

The individual contribution includes the market-data preparation workflow, feature engineering, behaviour modelling, anomaly-aware comparison, explainability calculations, interaction and consistency analysis, runtime integration, output safeguards, test artefacts, and technical documentation for the market component.

The contribution also includes translating the model output into dashboard-ready structures so that a user can inspect the current price, expected path, deviation evidence, influential factors, risk context, and limitations within the integrated project.

1.9 Thesis structure

Chapter 1 introduces the project, the market component, the problem, research gap, objectives, questions, scope, and contribution. Chapter 2 reviews financial time-series modelling, anomaly detection, explainable AI, and investor-facing analytics. Chapter 3 describes the research and evaluation methodology. Chapter 4 presents the requirements and solution design. Chapter 5 explains implementation, integration, testing, and operation. Chapter 6 reports the technical and validation results. Chapter 7 discusses the findings and limitations. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the thesis and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review focused on research and practice related to stock-price forecasting, financial time-series analysis, anomaly detection, explainable artificial intelligence, feature interaction analysis, and investor-oriented visualisation. The review also considered the distinction between predictive association and causal explanation, because this distinction is important when presenting market evidence to non-specialist users.

The review was used to identify design requirements rather than to claim that one algorithm is universally superior. The resulting component therefore uses a modular structure in which models, features, anomaly rules, and explanation methods can be evaluated or replaced independently.

2.2 Domain and theoretical foundations

Market modelling treats a share price as a time-dependent observation influenced by prior prices, trading activity, volatility, liquidity, and wider market conditions. Useful features may include lagged returns, rolling statistics, price ranges, volume changes, and other indicators that summarise behaviour without assuming that the market is fully predictable.

Explainable AI provides a bridge between a model output and a human interpretation. Feature attribution estimates how inputs contributed to a prediction,

while interaction analysis examines whether the contribution of one variable changes in the presence of another. In this thesis, explanations are treated as model evidence and are presented with limitations rather than as independent causal claims.

Anomaly detection compares an observed value or behaviour pattern with an expected baseline. A deviation becomes more informative when the system records the date, magnitude, direction, score, threshold, liquidity context, and relevant external events.

2.3 Related research

Prior financial forecasting research includes statistical time-series models, regression approaches, machine-learning ensembles, and deep-learning methods. These approaches differ in their assumptions, data requirements, interpretability, and sensitivity to changing market regimes. Studies of explainable financial models show that model transparency is especially important when predictions influence decisions.

Research on anomaly detection commonly defines unusual observations by distance from an expected value, residual behaviour, density, or a learned representation. Research on explainability adds attribution and interaction methods, but the practical presentation of these outputs to emerging-market retail investors remains underdeveloped. The present work combines these strands in a single operational workflow.

2.4 Existing systems products and techniques

Existing financial platforms provide charts, indicators, news, and sometimes forecasts, but they often require users to understand the meaning and limitations of each signal. Generic dashboards may also separate data sources without showing how evidence should be combined. Global products are not necessarily configured for the data availability, terminology, and investor needs of the CSE.

The selected approach extends conventional dashboards by linking the predicted path to anomaly evidence and explanations. It keeps the interface investor-oriented while retaining technical details that support review, such as model comparison, deviation history, factor movement, and explanation caveats.

2.5 Gap synthesis and conceptual framework

The conceptual framework begins with validated historical market data and proceeds through preprocessing, feature engineering, model training, chronological testing, prediction, anomaly comparison, explanation, consistency analysis, and dashboard presentation. External context is connected as supporting evidence and not merged into a causal claim.

The framework is designed around traceability. Every displayed market statement should be linked to a value, date, calculation, model output, or explicit limitation. This makes the component suitable for integration with the other project modules and for later evaluation by a supervisor or technical reviewer.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a design-and-evaluate approach. A working market-analysis artefact was developed, tested using historical observations, integrated into the web application, and reviewed against functional and research requirements. The main independent choices are the feature set, modelling workflow, anomaly threshold, and explanation method. The dependent outcomes include forecast error, anomaly detection behaviour, explanation stability, and usability of the displayed evidence.

Chronological validation is used to reduce leakage from future observations. Training data precede validation and test observations, and the evaluation records the comparison with a simple unchanged-price baseline where applicable.

3.2 Data participants or study materials

The study materials are historical CSE market observations for the supported companies, including trade dates, open, high, low, close, adjusted close, and volume fields. The integrated system also uses current quote refreshes, selected company metadata, external market indicators, and dated news context when available.

No human participant data is required for the core modelling experiment. User accounts and uploaded reports are handled by the surrounding application and are protected through authentication and controlled access. The research artefacts

contain model outputs and validation records rather than private investor information.

3.3 Experimental or study procedure

The procedure starts by validating the stock symbol and loading the historical dataset. Missing or invalid observations are identified, dates are ordered, and the model features are prepared. Candidate modelling steps are trained on earlier observations and tested on later observations. The runtime then refreshes available stock quotes before a new analysis is created.

For each analysis, the component creates forecast horizons and compares the expected path with the latest actual price. An anomaly score and threshold determine whether a deviation is flagged. Explanation outputs identify influential factors, interactions, and consistency indicators. Finally, a public-safe result is returned to the dashboard with limitations and a non-advisory statement.

3.4 Measures and evaluation metrics

Forecast performance is assessed using error measures such as mean absolute error and comparison with an unchanged-price baseline. Directional or interval behaviour may be considered where supported by the artefacts. Anomaly evaluation records detection counts, thresholds, deviations, and dates rather than presenting a binary flag without context.

Explainability is assessed by inspecting whether the reported drivers correspond to available features, whether contribution signs are internally coherent, and whether the important drivers remain reasonably stable across evaluation windows. Software quality is evaluated through automated tests, API checks, validation of malformed inputs, and integration tests.

3.5 Data analysis

Data analysis combines descriptive statistics, chronological model evaluation, residual or deviation inspection, and explanation summaries. The output is interpreted at the level of evidence available for a selected company and period. Results are not generalised to every CSE stock or every future market regime.

The analysis also compares model output with a simple baseline. If a sophisticated estimate does not outperform the baseline in a recent test, the dashboard labels the estimate as requiring extra caution. This safeguard prevents technical complexity from being mistaken for demonstrated predictive superiority.

3.6 Ethics privacy security and responsible AI

The component is designed as an informational and educational support tool rather than an automated financial adviser. The user interface states that estimates are uncertain and not financial advice. Explanations use cautious language, and the system distinguishes association from proof of causation.

Credentials are supplied through environment variables and are not included in generated outputs. Access to analyses and uploaded reports is controlled by authenticated routes. Data validation, file-size checks, temporary analysis expiry, and safe warning messages reduce unnecessary exposure of sensitive information or internal implementation details.

3.7 Reproducibility and research artefacts

The repository separates source datasets, locked or generated model artefacts, validation results, runtime adapters, and application services. Configuration values are externalised, and the research component can be invoked from the server using the configured Python environment. This separation supports repeatable testing and makes it possible to update one modelling stage without rewriting the entire application.

The main reproducibility risks are changes in upstream market data, external news availability, dependency versions, and model randomness. These risks are reduced through stored evaluation artefacts, explicit configuration, chronological validation, and documented limitations.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements include secure access, selection of a supported CSE stock, retrieval of historical data, calculation of market features, generation of forecast

paths, detection of unusual movements, explanation of influential factors, comparison of expected and actual values, and display of limitations. The component must also provide a stable output structure for the integrated dashboard.

Non-functional requirements include reliability when external data is incomplete, maintainability through modular services, acceptable response time for a research prototype, security of user and report data, traceability of calculations, and usability for investors who may not be familiar with machine-learning terminology.

4.2 Component architecture within the complete solution

The complete system follows a modular architecture. The React client provides the user workspace and insight presentation. The Express server authenticates users, manages reports, coordinates analysis stages, and returns public-safe results. The market component runs its research-supported Python workflow, while other services process reports, news and sentiment, and external-market risk.

The market component receives a stock symbol and relevant data paths, then returns structured market evidence. The integration layer combines this evidence with the other components only after each stage has completed or reported a limitation. The architecture allows the market module to be evaluated independently while keeping the user experience unified.

4.3 Design alternatives and decisions

A purely rule-based dashboard was rejected because it would not adapt well to changing price behaviour or provide a learned expected path. A black-box predictor without explanations was rejected because it would not satisfy the transparency requirement. A single global model was also considered unsuitable because supported CSE stocks can differ in liquidity and behaviour.

The selected design uses a reproducible modelling pipeline with explicit features, chronological testing, anomaly-aware comparison, explainability outputs, and a cautious baseline check. This provides a practical balance between quantitative analysis and reviewability for the current project scope.

4.4 Detailed design

The data flow is: stock selection, data refresh, historical data loading, cleaning, feature engineering, model execution, forecast generation, anomaly scoring, explanation generation, consistency assessment, public-output filtering, and dashboard rendering. Each stage records enough information for the next stage while excluding internal paths or sensitive implementation details from the public response.

The anomaly design stores the actual value, expected value, signed deviation, deviation percentage, score, threshold, observation window, and detected event history. The explanation design stores factor labels, latest values, recent changes, business meaning, possible implications if a factor rises or falls, and an association disclaimer.

4.5 Standards and professional practice

The implementation follows separation of concerns, environment-based configuration, input validation, authenticated access control, explicit error handling, and automated testing. Research outputs are kept separate from the web presentation layer so that the evidence can be inspected independently of the interface.

The component follows responsible-AI practice by exposing limitations, avoiding unsupported causal language, preserving unavailable values instead of inventing them, and presenting the result as non-advisory information. The formatting and documentation approach follows the supplied SLIIT dissertation guidance, including A4 layout, Times New Roman body text, structured headings, and consistent references.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The integrated application uses a React and Vite client, an Express and Node.js server, and MongoDB for users, stock records, reports, and temporary analysis runs. The market modelling component uses Python with numerical, machine-learning, explainability, and data-processing libraries. The runtime communicates through authenticated API routes and structured JSON results.

Development configuration is kept outside source code. The local runtime uses a Python 3.11 virtual environment for the research dependencies, while the web client and server use their respective package manifests. This arrangement supports local development and allows the component to be moved to a managed deployment environment later.

5.2 Component implementation

The implementation loads validated price history, derives trend and volatility features, trains or invokes the configured market-analysis workflow, and stores the resulting forecast and quality information. The public output removes internal source paths and model-only details while retaining the evidence needed to understand the result.

The anomaly stage compares the latest actual price with the expected value and maintains a recent history of detected events. The explanation stage describes the most influential drivers and their movements. The consistency stage compares driver behaviour over time so that a single attribution is not presented as permanently reliable.

5.3 Integration with the other components

The server starts the market, report, news and sentiment, and market-risk stages as coordinated analysis work. The final integration layer combines their public-safe outputs into a single insight. The market component remains responsible for quantitative market evidence, while the report and news components provide complementary context rather than being treated as replacements for market validation.

The dashboard presents the result in sections for market behaviour, unusual movement, report evidence, news and external risk, and the combined explanation. The output-language control requests the analysis in English, Sinhala, or Tamil while preserving the numeric values and URLs needed for interpretation.

5.4 Verification and software testing

Verification covers data preparation, model invocation, output shape, anomaly thresholds, missing-data handling, public-output filtering, and API integration. The web application was checked using the client lint and production build commands, while the server modules were checked for syntax and the local health endpoint was verified after the database connection was established.

Testing also considers failure paths. If an external data source or optional explanation service is unavailable, the application returns a controlled limitation rather than exposing a stack trace or presenting an unverified conclusion. If the advanced estimate does not beat the unchanged-price comparison, the result is labelled with additional caution.

5.5 Deployment and operation

For local operation, MongoDB is started as a local service, the server loads environment variables and starts on port 5001, and the Vite client starts on port 5173. The server first connects to MongoDB and then exposes the health route and authenticated application routes. A production deployment would require managed database access, secret management, HTTPS, protected file storage, monitoring, and controlled Python runtime packaging.

A new analysis refreshes available quote data, validates the uploaded report through the surrounding services, runs the analysis stages, stores the result temporarily, and returns a dashboard-ready response. Analysis records expire after the configured period, which limits long-term storage of uploaded-report results.

6 RESULTS AND VALIDATION

6.1 Research results

The developed component produces a complete market evidence object for supported stocks. The object includes the current market value, historical behaviour, forecast horizons, quality indicators, anomaly status, deviation history, influential factors, and limitations. This is a stronger research output than a single predicted price because it gives the user several ways to inspect the evidence.

Chronological validation and baseline comparison are used to avoid claiming success from an in-sample fit. The recorded research artefacts show that market forecasts must be interpreted cautiously because performance can vary across stocks and time windows. The system therefore keeps the baseline result and uncertainty visible in the product output.

6.2 Product and technical results

The component integrates successfully with the Express analysis workflow and the React dashboard. The health endpoint confirms server availability, authenticated analysis routes return structured responses, and the market output is rendered in separate sections for trend, forecast, anomaly, and explanatory context.

The implementation also supports multilingual output for generated analysis text. English remains the default, while Sinhala and Tamil can be selected from the shared header control. Translation is applied to analysis values on request and falls back to the original text if the external translation service is unavailable.

6.3 User or stakeholder validation

The current validation is primarily technical and scenario-based rather than a completed controlled human-user study. The dashboard structure was designed around the needs of retail investors, analysts, administrators, and researchers identified in the proposal. The plain-language sections, visible caveats, and separated evidence cards are intended to reduce the cognitive burden of interpreting the output.

A formal user study should measure comprehension, trust, time to locate evidence, ability to identify uncertainty, and whether users mistakenly interpret the output as investment advice. Those measures are outside the completed prototype validation and are retained as future work.

6.4 Integration results

The integrated workflow brings market behaviour, report evidence, news context, and external-market risk into one analysis view. The market component contributes the quantitative baseline and unusual-movement evidence, while the other stages add

context that can support or qualify the interpretation. The final result keeps the distinction between simultaneous evidence and proven cause.

The integration also demonstrates graceful degradation. A missing or incomplete stage is represented through a warning or unavailable state instead of silently being treated as evidence. This is important for a system that depends on multiple data sources with different availability and reliability.

6.5 Summary of findings

The component meets the main technical objectives of producing a reproducible, explainable, and anomaly-aware market output for the CSE. It improves interpretability by connecting estimates to factor contributions, deviations, and limitations. It does not remove the uncertainty inherent in market forecasting, and its results should remain subject to independent review against the underlying data.

7 DISCUSSION

7.1 Interpretation of findings

The most important finding is that a useful market insight is more than a forecast number. Investors need to know the expected path, the size and direction of unusual movement, the influential factors, the quality of the estimate, and what the model cannot establish. The component addresses this by treating explanation and limitation as first-class outputs.

The findings also confirm that the value of explainability depends on careful language. A factor contribution can describe how a model used a variable, but it cannot by itself prove that the variable caused a market movement.

7.2 Comparison with literature baselines and existing solutions

Compared with a chart-only interface, the component supplies explicit expected values, anomaly thresholds, and explanatory drivers. Compared with a black-box forecast, it makes the evidence path more visible. Compared with a simple rule-based signal, it can learn historical behaviour and evaluate its estimates chronologically.

The main trade-off is complexity. More stages create more opportunities for data quality problems and require careful communication. The chosen design accepts this trade-off because the project goal is understandable, evidence-aware insight rather than a single opaque score.

7.3 Research contribution

The research contribution is a practical framework for applying market modelling and explainable AI to CSE data. It combines expected behaviour, anomaly-aware comparison, feature attribution, interactions, and consistency checking, then evaluates the resulting evidence through a reproducible runtime and stored artefacts.

The work also contributes an explicit distinction between model association and causal explanation in the presentation of market evidence. This distinction is important for responsible use in a financial setting.

7.4 Product contribution

The product contribution is an operational market-analysis module that can be used inside the integrated dashboard. It converts technical calculations into an investor-oriented narrative without hiding the supporting values, dates, thresholds, or limitations. The module is designed to be extended with additional stocks, features, and validation artefacts.

7.5 Limitations and threats to validity

The results are limited by the available CSE data, supported stocks, historical window, data-source changes, and the non-stationary nature of financial markets. A model that performs acceptably in one period may not perform similarly after a structural change, liquidity shock, policy event, or unusual external condition.

Anomaly thresholds can produce false positives or false negatives. Explainability methods describe the fitted model and may be sensitive to correlated features. External factors may overlap with stock movements without causing them. These limitations mean that the component should support human review rather than automate a decision.

7.6 Implications

For developers, the findings support modular pipelines, chronological testing, baseline comparison, and public output safeguards. For investors, the component can help locate and interpret evidence but should not replace the original market records, company disclosures, or professional advice. For future researchers, the project provides a foundation for studying explainability and anomaly detection in a smaller emerging-market dataset.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The value proposition is a clearer and more transparent way to interpret CSE price behaviour. Retail investors are the primary users, especially beginner and intermediate investors who need explanations rather than isolated indicators. Financial analysts, brokerage firms, educators, researchers, and system administrators are additional stakeholders identified in the proposal.

8.2 Market and competitive position

Global platforms such as charting and market-data services provide valuable information but often expect users to interpret signals independently. The proposed product differentiates itself through its focus on the CSE, its combination of market, report, news, and risk evidence, and its emphasis on explanation and limitations. The competitive position should be evaluated through user comprehension and trust as well as technical functionality.

8.3 Business or sustainability model

A possible sustainability model is a tiered software-as-a-service offering. A basic tier could provide limited market views, while premium or institutional access could provide deeper explainability, report analysis, historical comparisons, and API access. Educational and research deployments could use cached artefacts and limited data windows to control computational cost.

Operational sustainability depends on maintaining data sources, monitoring model quality, updating dependencies, managing uploaded documents securely, and reviewing whether the explanations remain meaningful as market conditions change.

8.4 Scalability intellectual property and compliance

Scalability requires asynchronous job handling, caching, queue-based analysis, stronger storage controls, and a larger validation portfolio. The modular architecture supports adding stocks, indicators, anomaly methods, and deployment capacity without replacing the entire platform.

The source code, interface, workflow, integration design, and generated research artefacts may be protected as project intellectual property. Any external data must be used according to its licence and terms. Compliance also requires clear disclaimers, audit logs, access control, and a process for correcting unsupported outputs.

8.5 Contribution to the Sustainable Development Goals

The project can support SDG 4 by improving financial and digital literacy, SDG 8 by making business and market information more accessible, and SDG 9 through applied digital infrastructure for an emerging market. These benefits depend on keeping the system accessible, transparent, non-advisory, and honest about uncertainty.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis developed the Market Modeling and Explainable AI Insight Generation component for an integrated CSE stock insight platform. The component combines historical data preparation, expected behaviour modelling, anomaly-aware comparison, feature explanation, interaction analysis, and consistency checks in a single runtime. Its output is designed to be understandable without removing the technical evidence needed for review.

9.2 Achievement of objectives

The work achieved the objectives of establishing a reproducible market-data workflow, preparing modelling features, generating forecast evidence, detecting unusual deviations, exposing influential factors, integrating the result with the complete application, and validating the main runtime paths. The evaluation also made the limitations of short-horizon market estimation visible rather than hiding them behind a single score.

9.3 Individual research and product contributions

The research contribution is the combination of anomaly-aware market modelling and explainability for a CSE-focused investor workflow. The product contribution is the tested component and dashboard integration that presents price paths, deviations, drivers, context, and uncertainty together. The work also contributes multilingual analysis output support through Sinhala and Tamil translation of generated analysis content.

9.4 Future directions

Future work should extend the supported stock universe, incorporate corporate-action adjustments and richer market context, improve anomaly validation, compare additional model families, and conduct a larger rolling-origin evaluation. Further work should also evaluate explanation stability with human reviewers, add formal user testing, improve Sinhala and Tamil terminology quality, introduce stronger offline translation options, and study how users interpret uncertainty and non-advisory warnings.

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GLOSSARY

Anomaly: An observation whose measured behaviour differs materially from the expected baseline.

Explainable AI: Methods that show how model inputs contribute to an output.

Feature attribution: An estimate of the contribution of an input feature to a model prediction.

Forecast interval: A range expressing uncertainty around an estimated future value.

Liquidity: The degree to which an asset can be traded without a large price effect.

Model association: A relationship identified by a model that does not establish causation.

APPENDICES

Appendix A Research instruments or extended results

The supporting artefacts include the market-data preparation files, chronological validation outputs, forecast comparison tables, anomaly observations, feature contribution summaries, interaction results, and integration test records. These artefacts are stored in the project repository and should be attached or referenced according to supervisor instructions.

Appendix B Academic integrity and AI use disclosure

Tools used in preparing this thesis included Microsoft Word-compatible document tooling, Python data-processing and machine-learning libraries, the project repository, and AI-assisted drafting support. AI-assisted text was treated as draft material and checked against the supplied proposal, implementation behaviour, and research artefacts. The author remains responsible for the accuracy, originality, citations, code, data, analyses, and conclusions.